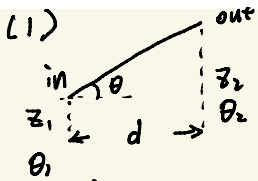


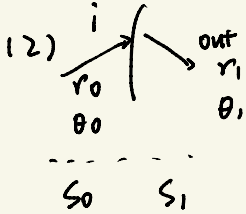
❖ Derive the ray matrices for cases (1) to (5) on next page.



$$z_2 = z_1 + d \cdot \tan \theta = z_1 + d \cdot \theta$$

$$\theta = \theta_1 = \theta_2$$

$$\Rightarrow \begin{bmatrix} z_2 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} 1 & d \\ 0 & 1 \end{bmatrix} \begin{bmatrix} z_1 \\ \theta_1 \end{bmatrix} \quad \leadsto \quad \underline{\underline{\begin{bmatrix} 1 & d \\ 0 & 1 \end{bmatrix}}}$$



$$r_1 = r_0$$

$$\frac{1}{s_0} + \frac{1}{s_1} = \frac{1}{f} \Rightarrow \frac{\theta_0}{r_0} - \frac{\theta_1}{r_1} = \frac{1}{f}$$

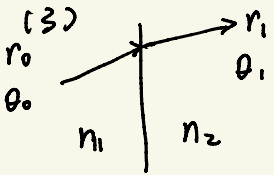
$$\theta_0 = \frac{r_0}{s_0}$$

$$\theta_1 = \frac{r_1}{s_1}$$

$$\Rightarrow r_1 = r_0$$

$$\theta_0 - \theta_1 = \frac{1}{f} r_0 \Rightarrow \theta_1 = -\frac{1}{f} r_0 + \theta_0$$

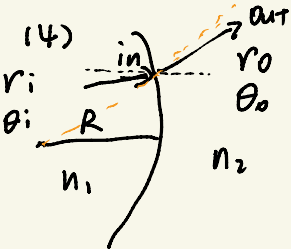
$$\Rightarrow \begin{bmatrix} r_1 \\ \theta_1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -\frac{1}{f} & 1 \end{bmatrix} \begin{bmatrix} r_0 \\ \theta_0 \end{bmatrix} \quad \leadsto \quad \underline{\underline{\begin{bmatrix} 1 & 0 \\ -\frac{1}{f} & 1 \end{bmatrix}}}$$



$$r_0 = r_1$$

$$n_1 \cdot \theta_0 = n_2 \theta_1 \Rightarrow \theta_1 = \frac{n_1}{n_2} \theta_0$$

$$\Rightarrow \begin{bmatrix} r_1 \\ \theta_1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & \frac{n_1}{n_2} \end{bmatrix} \begin{bmatrix} r_0 \\ \theta_0 \end{bmatrix} \quad \leadsto \quad \underline{\underline{\begin{bmatrix} 1 & 0 \\ 0 & \frac{n_1}{n_2} \end{bmatrix}}}$$



$$r_0 = r_i$$

$$\theta_i + \theta_1 = \theta_0 + \theta_2 = \frac{r_0}{R} \Rightarrow n_1 \left(\frac{r_0}{R} - \theta_i \right) = n_2 \left(\frac{r_0}{R} - \theta_0 \right)$$

$$n_1 \cdot \theta_1 = n_2 \theta_2$$

$$\Rightarrow \theta_0 = \frac{n_1}{n_2} \theta_i - \frac{n_1}{n_2} \cdot \frac{r_0}{R} + \frac{r_0}{R}$$

$$\Rightarrow \begin{bmatrix} r_0 \\ \theta_0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ \frac{n_2 - n_1}{n_2 R} & \frac{n_1}{n_2} \end{bmatrix} \begin{bmatrix} r_i \\ \theta_i \end{bmatrix} \quad \leadsto \quad \underline{\underline{\begin{bmatrix} 1 & 0 \\ \frac{n_2 - n_1}{n_2 R} & \frac{n_1}{n_2} \end{bmatrix}}}$$



$r_2 = r_1$
 $\theta_1 + \theta_2 = 2\left(\theta_1 - \frac{r_0}{R}\right)$
 $\Rightarrow \theta_2 = \theta_1 - 2\frac{r_0}{R}$
 $\Rightarrow \begin{bmatrix} r_2 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -\frac{2}{R} & 1 \end{bmatrix} \begin{bmatrix} r_1 \\ \theta_1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 \\ -\frac{2}{R} & 1 \end{bmatrix}$

❖ Ch2_No.3

Based on the matrix formulation for the light propagation, prove that the confocal cavity is a stable cavity, that is, any paraxial light can go back and forth within it multiple times, and it will form close-loop after two rounds.

$$R_1 = R_2 = R \Rightarrow T = \begin{bmatrix} 1 & R \\ 0 & 1 \end{bmatrix}$$

$$M = \begin{bmatrix} 1 & 0 \\ -\frac{2}{R} & 1 \end{bmatrix}$$

$$\Rightarrow M_{RT} = (M \cdot T) \cdot (M \cdot T)$$

$$= \begin{bmatrix} 1 & R \\ -\frac{2}{R} & -1 \end{bmatrix} \begin{bmatrix} 1 & R \\ -\frac{2}{R} & -1 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \Rightarrow -1 \leq \frac{A+D}{2} \leq 1 \Rightarrow \text{close loop}$$

$$M_{RT}^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

$$\text{Tr}(M_{RT}) \leq 2 \Rightarrow -1 \leq -1 \leq 1$$

$\Rightarrow$ stable cavity

❖ Ch2_No.5

The resonant cavity of the laser is composed of a convex mirror with a curvature of 1 m and a concave mirror with a curvature of 2 m, respectively. The working substance is 0.5 m long with refractive index of 1.52. Please determine the range of cavity length L to form a stable cavity.

$$L_{\text{eff}} = (L-d) + \frac{d}{n} = L - \frac{1}{2} \cdot \frac{0.52}{1.52}$$

$$\Rightarrow g_1 = 1 - \frac{L_{\text{eff}}}{R_1}$$

$$g_2 = 1 - \frac{L_{eff}}{R_2} \Rightarrow 0 \leq g_1, g_2 \leq 1$$

$$\Rightarrow 0 \leq (1 + L_{eff}) \left(1 - \frac{L_{eff}}{2}\right) \leq 1$$

$$\Rightarrow \begin{cases} L_{eff} (1 - L_{eff}) \leq 0 \\ 1 - \frac{L_{eff}}{2} \geq 0 \end{cases} \Rightarrow 1 \leq L_{eff} \leq 2$$

$$\Rightarrow 1 \leq L - \frac{1}{2} \cdot \frac{0.52}{1.52} \leq 2$$

$$\Rightarrow \frac{89}{76} \leq L \leq \frac{165}{76}$$

$$\Rightarrow 1.171m \leq L \leq 2.171m$$